TESS Stellar Analysis Suite

User and Research Manual

Version 1.0.0

Developed by

Javed Iqbal
Dr. Ahmed Waqas Zubairi
Dr. Shaukat Goderya

A desktop research tool for retrieving, processing, visualizing, and analyzing TESS light curves from a TIC identifier.

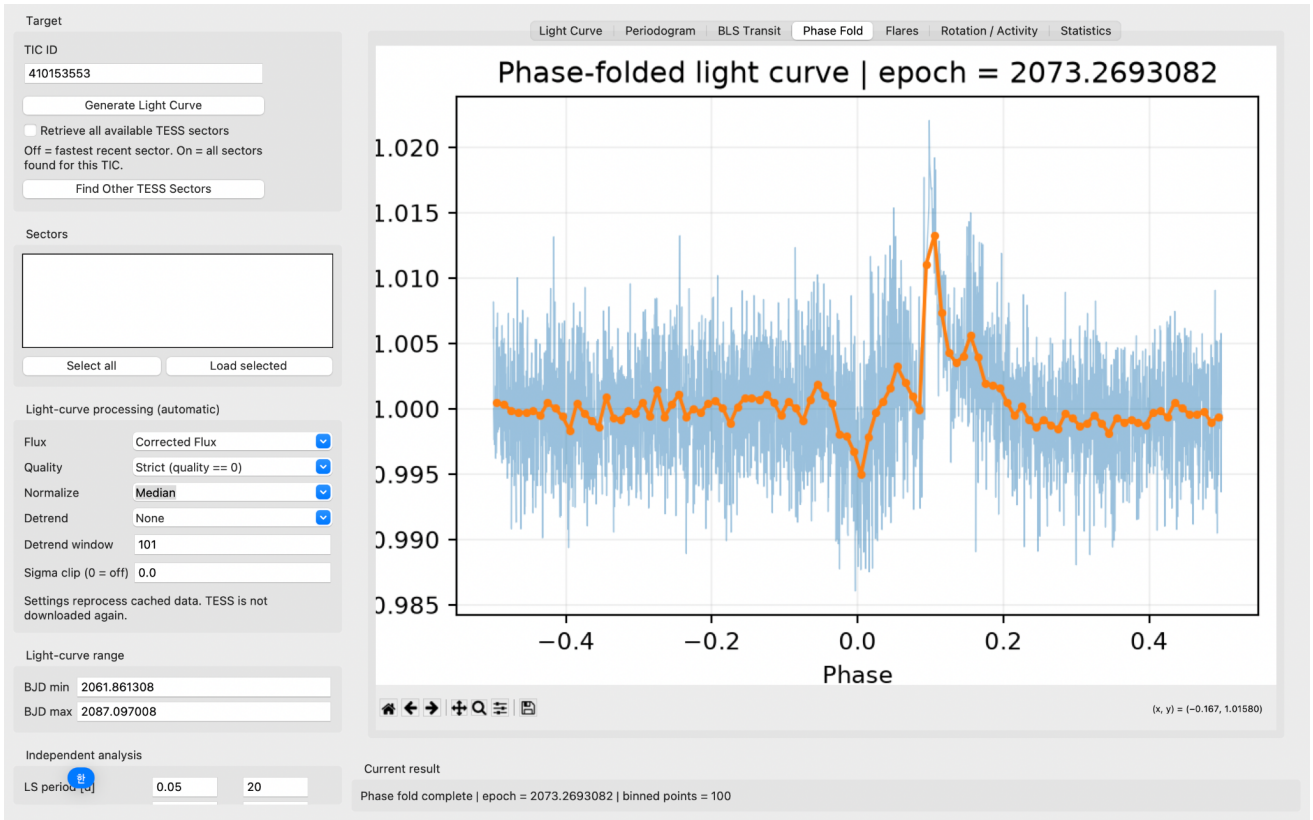


Figure 1. Example interface showing the analysis tabs and phase-folded light curve.

1. Purpose and Scope

TESS Stellar Analysis Suite is a desktop graphical application designed for exploratory and research-oriented analysis of TESS time-series photometry. A user supplies a TESS Input Catalog (TIC) identifier, retrieves available TESS observations through the Eleanor workflow, processes the resulting light curve, and then performs several independent analyses from the same processed data.

The suite is broader than a light-curve generator. It combines light-curve retrieval and processing with periodic-variability analysis, transit screening, phase folding, flare screening, stellar rotation/activity analysis, statistics, and data/figure export.

The software is intended to support research workflows and teaching. Its outputs are analysis products that require scientific interpretation and, where appropriate, follow-up validation.

2. Development and Acknowledgement

The project is developed by:

- Javed Iqbal
- Dr. Ahmed Waqas Zubairi
- Dr. Shaukat Goderya

The current implementation is built around the Eleanor package for TESS light-curve extraction and uses Astropy, Photutils, TensorFlow, Lightkurve, SciPy, NumPy, Pandas, and Matplotlib for data handling, photometry, time-series analysis, and visualization.

The internal Python module retains the name tess_lc_generator for compatibility with the current codebase. The public project name is TESS Stellar Analysis Suite.

3. What the Suite Can Do

Capability	Description
TESS data retrieval	Retrieve TESS observations from a TIC ID using Eleanor; use the fastest recent sector or search/retrieve all available sectors.
Light-curve processing	Select raw, corrected, or PCA flux; apply quality filtering, normalization, detrending, and optional sigma clipping.
Light-curve visualization	Display the processed time series and inspect a selected BJD range.
Lomb–Scargle periodogram	Search for periodic variability and report the strongest period, power, false-alarm probability, and additional peaks.
BLS transit screening	Search for transit-like periodic signals and report candidate period, depth, duration, power, and transit epoch.
Phase folding	Fold the light curve on a chosen period and epoch, with optional binning. A BLS candidate can be transferred directly.
Flare screening	Identify candidate positive excursions using robust statistics and report event timing, amplitude, duration, significance, and equivalent-duration information where applicable.
Rotation/activity	Search a rotation-oriented period range and inspect strong peaks, harmonics, and aliases.

Statistics	Summarize the processed time series, cadence, point count, flux distribution, and scatter/noise properties.
Export	Save processed light curves, FITS data, metadata, analysis results, and the current plot.
Performance	Use background processing and display decimation for smoother interaction with long TESS time series.
Automatic re-analysis	When processing settings change, cached data are reprocessed and previously activated analyses are recalculated.

4. System Requirements

- macOS, Windows, or Linux with a working Python/Conda installation. The supplied development environment was tested with Python 3.11.
- Internet access is required when TESS data or Eleanor metadata products must be downloaded.
- A modern desktop computer is recommended because TESS time series and some analyses can contain many thousands of cadences.
- The GUI uses Tkinter. On some Linux installations, the Tk package may need to be installed separately.

Known development environment:

Component	**Version used in the supplied project**
Python	3.11
NumPy	1.26.4
SciPy	>=1.11, <2
Astropy	6.1.7
Photutils	1.13.0
Eleanor	2.0.5
TensorFlow	2.16.2
Matplotlib	>=3.8, <4
Pandas	>=2.0, <3
Lightkurve	>=2.4, <3

5. Installation

The recommended method is to create a clean Conda environment so that the astronomy dependencies do not interfere with other projects.

```
conda create -n tess-analysis python=3.11
conda activate tess-analysis
```

From the project directory, install the supplied dependencies:

```
pip install -r requirements.txt
```

The supplied requirements file pins the dependency combination used during development, including Eleanor 2.0.5, Photutils 1.13.0, TensorFlow 2.16.2, Astropy 6.1.7, and NumPy 1.26.4.

If you already have a working environment, for example an environment named tess-lc, you can activate that environment instead of creating a new one.

```
conda activate tess-lc
```

6. Starting the Application

Open a terminal and move into the project directory:

```
cd ~/Downloads/tess-stellar-analysis-suite
conda activate tess-analysis
python run.py
```

The GUI opens as a desktop window. The current source may still display the legacy window title "TESS Light Curve Generator"; this does not change the broader project scope or functionality.

A successful launch may display non-fatal warnings from Lightkurve or TensorFlow. Warnings such as the optional oktopus/tpfmodel message do not by themselves mean that the application failed.

7. Basic Workflow: From TIC ID to Analysis

1. Enter the TIC ID in the Target section.
2. Click Generate Light Curve.
3. For the fastest workflow, leave Retrieve all available TESS sectors unchecked. The suite will use the fastest recent sector available for the target.
4. If a broader time baseline is required, enable Retrieve all available TESS sectors and use Find Other TESS Sectors or the sector controls as appropriate.
5. Wait for retrieval and processing to finish. The status area reports progress.
6. Inspect the Light Curve tab first.
7. Adjust processing settings if necessary. Changes are applied automatically using cached data when possible, so TESS data do not need to be downloaded again merely because a processing option changed.
8. Open the analysis tabs individually: Periodogram, BLS Transit, Phase Fold, Flares, Rotation / Activity, and Statistics.
9. Export the processed data and analysis products needed for your research record.

8. Target and Sector Controls

TIC ID

Enter the numerical TESS Input Catalog identifier. The application validates the identifier before attempting retrieval.

Generate Light Curve

Starts the main retrieval and processing workflow.

Retrieve all available TESS sectors

When off, the application prioritizes a fast recent-sector workflow. When on, it searches for available sectors and retrieves the broader set.

Find Other TESS Sectors

Searches for additional TESS sectors containing the target. This is useful when a target has observations separated in time.

Sector list

Sectors can be selected and loaded. Select all selects all displayed sectors; Load selected loads the chosen sectors.

9. Light-Curve Processing

Processing is automatic. Changing a processing setting reprocesses the cached raw data rather than downloading TESS data again.

Setting	Available choices	Purpose
Flux	Raw Flux; Corrected Flux; PCA Flux	Choose the flux product used to build the working light curve.
Quality	Strict (quality == 0); Remove common artifacts; Allow flagged cadences	Control which cadences are retained.
Normalize	Median; Mean; Percent deviation; None	Put the flux on a convenient relative scale.
Detrend	None; Savitzky-Golay; Median filter	Remove slow instrumental or baseline trends.
Detrend window	Numeric window	Controls the smoothing/filter scale used by the selected detrending method.
Sigma clip	Numeric; 0 = off	Reject strong outliers according to the selected sigma threshold.

Use caution when detrending. A filter window that is too aggressive can remove real astrophysical variability or distort shallow transit signals. For transit work, compare multiple reasonable processing choices.

10. Light-Curve Range

BJD min and BJD max define the time interval used for the displayed/analysis window. The application preserves the selected range when processing settings are changed where possible.

The plotted x-axis is expressed as BJD − 2457000 in the light-curve visualization. When using an epoch in another analysis, use the value reported by the application and keep the time convention consistent.

11. Light Curve Tab

The Light Curve tab is the primary diagnostic view. It shows the processed time series and provides the basis for all subsequent analyses.

- Check for obvious gaps, discontinuities, outliers, and instrumental-looking structures.
- Compare the visual appearance before and after detrending or sigma clipping.
- Use the BJD range controls to inspect a specific event or time interval.
- Do not interpret every short excursion as an astrophysical signal; TESS systematics and residual artifacts can remain.

12. Lomb–Scargle Periodogram

The Periodogram analysis uses a Lomb–Scargle search over a user-defined period range. The default range in the GUI is 0.05–20 days.

The output includes the strongest period, its power, a false-alarm probability estimate, and additional strong peaks. Multiple peaks are important because harmonics and aliases can occur.

Independent controls:

Parameter	Meaning
LS period min [d]	Lower period limit for the Lomb–Scargle search.
LS period max [d]	Upper period limit for the search.

Recommended interpretation:

- A strong periodogram peak is evidence for periodic variability, not automatically a rotation period or planet.
- Inspect nearby peaks and harmonics.
- Check whether the signal persists under reasonable processing choices.
- For stellar rotation, compare the candidate with the expected activity timescale and other diagnostics.

13. BLS Transit Screening

The BLS Transit analysis uses Box Least Squares to search for periodic, transit-like decreases in flux. The default GUI search range is 0.5–20 days.

Reported quantity	Meaning
Candidate period	Period associated with the strongest BLS signal.
Power	Relative strength of the BLS signal within the searched period range.
Depth	Estimated transit-like flux decrement.
Duration	Estimated transit duration.
Transit epoch	Reference transit time used for phase folding.

A BLS result is a candidate screening result, not an exoplanet confirmation. Candidate validation should include inspection of individual transits, odd/even events, data gaps, nearby contamination, centroid information where available, stellar parameters, and independent photometry or follow-up observations as appropriate.

When a BLS candidate is produced, the application can transfer the candidate period and epoch directly to Phase Fold for visual inspection.

14. Phase Fold

Phase folding converts a time series into orbital or rotational phase using a selected period and epoch. The suite can use a period/epoch supplied manually or inherited from a BLS candidate.

Parameter	Purpose
Phase period [d]	Period used for folding.
Phase epoch	Reference epoch used to define phase zero.
Phase bin [cycles]	Optional phase-bin size used to produce a binned representation.

The plot displays the phase-folded data and an optional binned representation. Binning is a visualization aid and should not replace inspection of the unbinned data when evaluating shallow or short-duration signals.

15. Flare Screening

The Flares analysis searches for positive flux excursions using robust baseline/scatter estimates. It reports candidate events rather than declaring confirmed stellar flares.

Parameter	Default	Purpose
Flare threshold [σ]	5.0	Required significance threshold above the robust baseline.
Flare min points	2	Minimum number of points contributing to a candidate event.

Reported event information can include event time, amplitude, duration, significance, and equivalent-duration information where applicable.

For scientific flare studies, candidates should be visually inspected and checked against data gaps, cosmic-ray-like outliers, instrumental artifacts, and the chosen detrending method. A physically meaningful flare-energy calculation additionally requires appropriate stellar parameters and flux calibration.

16. Rotation / Activity

Rotation / Activity performs a Lomb–Scargle search over a rotation-oriented period range. The default range is 0.2–30 days.

Parameter	Purpose
Rotation period min [d]	Lower bound of the activity/rotation search.
Rotation period max [d]	Upper bound of the search.

The result reports a leading activity-period candidate and additional peaks. Inspect harmonics and aliases before adopting a rotation period. For active stars, spot evolution and changing amplitudes can also make the periodogram more complex.

17. Statistics

The Statistics tab summarizes the current processed light curve, including basic time-series and flux statistics, cadence/point-count information, and scatter/noise-related quantities available from the current data.

Use these values as descriptive diagnostics. They are not substitutes for a complete photometric uncertainty model or a publication-specific noise analysis.

18. Exporting Results

Export option	Use
Export CSV	Save the processed light curve in a tabular form suitable for further analysis.
Export FITS	Save the processed light curve in an astronomy-friendly FITS format.
Export metadata JSON	Save target/processing metadata for reproducibility.
Export analysis JSON	Save analysis results and parameters for the current session.
Save current plot	Save the currently displayed analysis figure.

For research projects, keep the exported data together with the analysis JSON and a record of the processing parameters used to produce the result.

19. Automatic Updating and Independent Analyses

The suite separates the main processed light curve from the individual analyses. This allows Periodogram, BLS, Phase Fold, Flares, and Rotation / Activity to be inspected independently.

After an analysis has been run, changing its parameters can trigger an automatic recalculation. More importantly, changing the main light-curve processing settings reprocesses the cached data and recalculates analyses that have already been activated.

This design prevents a user from accidentally assuming that an old periodogram or BLS result still corresponds to a newly processed light curve.

20. Recommended Research Workflow

10. Retrieve the target by TIC ID.
11. Start with Corrected Flux, strict quality filtering, median normalization, and no detrending.
12. Inspect the complete light curve for gaps, jumps, and obvious artifacts.
13. Test a reasonable detrending choice if long-term baseline structure is present.
14. Run the Lomb–Scargle periodogram to identify periodic variability.
15. Run BLS over a physically motivated period range if transit screening is the scientific goal.
16. Inspect the BLS candidate in Phase Fold using the unbinned data before relying on the binned representation.
17. Run Flare screening if stellar activity is relevant, and visually inspect candidate events.
18. Run Rotation / Activity over an appropriate period range and compare the candidate with the general light-curve morphology.
19. Repeat key analyses under at least one alternative reasonable processing configuration when the scientific conclusion depends strongly on detrending or clipping.
20. Export the processed data, analysis results, and figures used in the scientific interpretation.

For publication-quality work, retain the raw/retrieved-data provenance and record the exact target, sector(s), processing settings, analysis ranges, and software environment.

21. Troubleshooting

Problem	Recommended action
The program does not start and reports a Photutils import error.	Eleanor 2.0.5 expects the Photutils API used by the supplied environment. Use the supplied requirements file or install Photutils 1.13.0 rather than a newer incompatible release.
TensorFlow is missing.	Install the supplied TensorFlow version with the requirements file. The tested environment uses TensorFlow 2.16.2.
Astropy fails after installing TensorFlow.	Avoid mixing incompatible NumPy/Astropy versions. Recreate the clean Python 3.11 environment and install the supplied requirements together.
TESS data are slow to retrieve.	The first retrieval for a sector can require TESSCut/Eleanor metadata downloads. Cached data and subsequent retrievals can be faster.
A target has no Eleanor postcard.	The suite can fall back to TESSCut. This may require downloading a target cutout and can be slower than a cached Eleanor postcard workflow.
A plot appears empty.	First verify that a light curve has been generated. Then check the selected BJD range and processing settings. Analysis tabs operate on the current processed light curve.
A BLS signal looks suspicious.	Treat it as a candidate only. Inspect individual events, the phase-folded light curve, data gaps, systematics, and stellar/field contamination before drawing conclusions.
The GUI appears busy during	Large TESS datasets can require substantial computation or

retrieval or analysis.	network transfer. Background processing is used where possible, but initial retrievals can still take time.

22. Scientific Limitations and Good Practice

- A detected periodic signal is not automatically a planet, rotation period, or other astrophysical phenomenon.
- BLS identifies transit-like periodic structure; it does not validate planetary nature.
- Periodogram peaks can arise from harmonics, aliases, spacecraft/systematic effects, or evolving stellar variability.
- Flare candidates require visual and astrophysical vetting.
- Detrending and sigma clipping can change the appearance and detectability of astrophysical signals.
- TESS sector gaps, cadence changes, and instrumental systematics should be considered when comparing signals across sectors.
- Results should be reproduced from saved processing settings and exported data before being used in a publication.

23. Suggested Citation / Acknowledgement Text

When distributing or using this software in a research or teaching context, the following acknowledgement may be used:

> *"We used the TESS Stellar Analysis Suite, developed by Javed Iqbal, Dr. Ahmad Waqas Zubari, and Dr. Shaukat Goderya for retrieval, processing, and analysis of TESS light curves."*

24. Project Structure

```
tess-stellar-analysis-suite/
├── run.py
├── requirements.txt
├── LICENSE
├── VERSION
└── tess_lc_generator/
    ├── __init__.py
    ├── app.py
    └── core.py
```

The current project package is a desktop GUI application. The manual describes the v1.0.0 project package supplied with this document.

25. Quick Reference

Task	Action
Launch	conda activate tess-analysis; python run.py
New target	Enter TIC ID → Generate Light Curve

More sectors	Enable Retrieve all available TESS sectors or use Find Other TESS Sectors
Change processing	Change Flux / Quality / Normalize / Detrend / window / sigma clip; processing updates automatically
Period search	Periodogram → set LS range → Run Periodogram
Transit screening	BLS Transit → set BLS range → Run BLS
Inspect candidate	Phase Fold → enter period/epoch or use BLS candidate
Flare screening	Set threshold/min points → Screen Flares
Rotation search	Set rotation range → Rotation / Activity
Summary	Statistics
Save data	Export CSV or FITS
Save provenance	Export metadata JSON and analysis JSON
Save figure	Save current plot

26. Final Note for Users

The TESS Stellar Analysis Suite is intended to make a complete first-pass time-series analysis workflow accessible from a single desktop interface. It should be used as a research assistant rather than as an automatic scientific decision maker. The strongest results come from combining the software outputs with visual inspection, knowledge of the target and observing conditions, appropriate statistical tests, and independent validation.